

# Assignment-19.1

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**Batch – 12**

## **Task 1 – Python to Java Conversion**

Task:

Translate a simple Python class into Java using AI assistance.

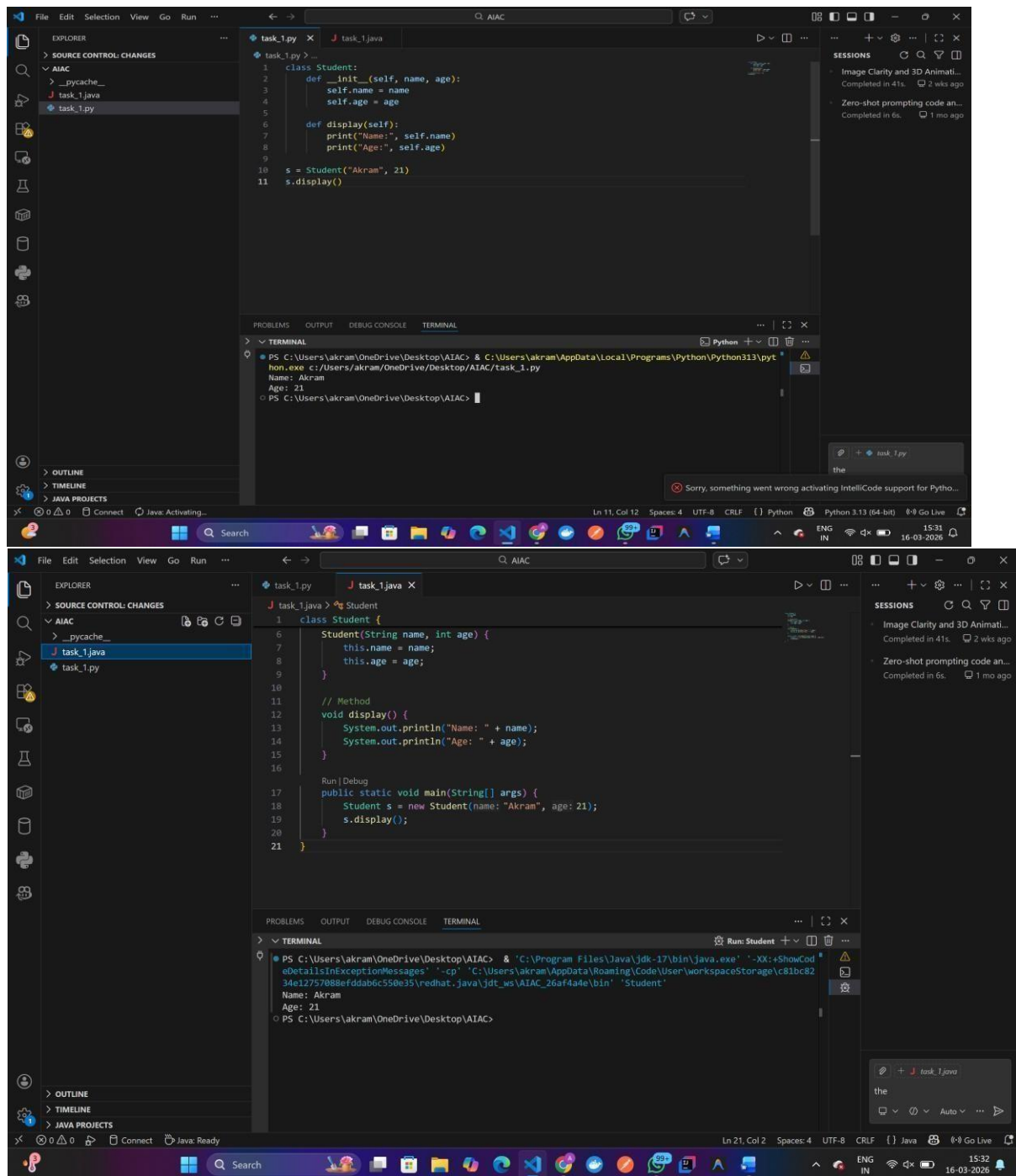
Instructions:

- Prompt AI to convert a Python class with attributes and methods into equivalent Java code.
- Verify that constructors, method signatures, and data types are properly handled.
- Run both versions (Python & Java) to compare expected outputs.

Expected Output:

- Equivalent working Java class translated from Python.

CODE:



## Task 2 – C++ to Python Function Conversion

Task:

Convert a C++ function for factorial calculation into Python.

Instructions:

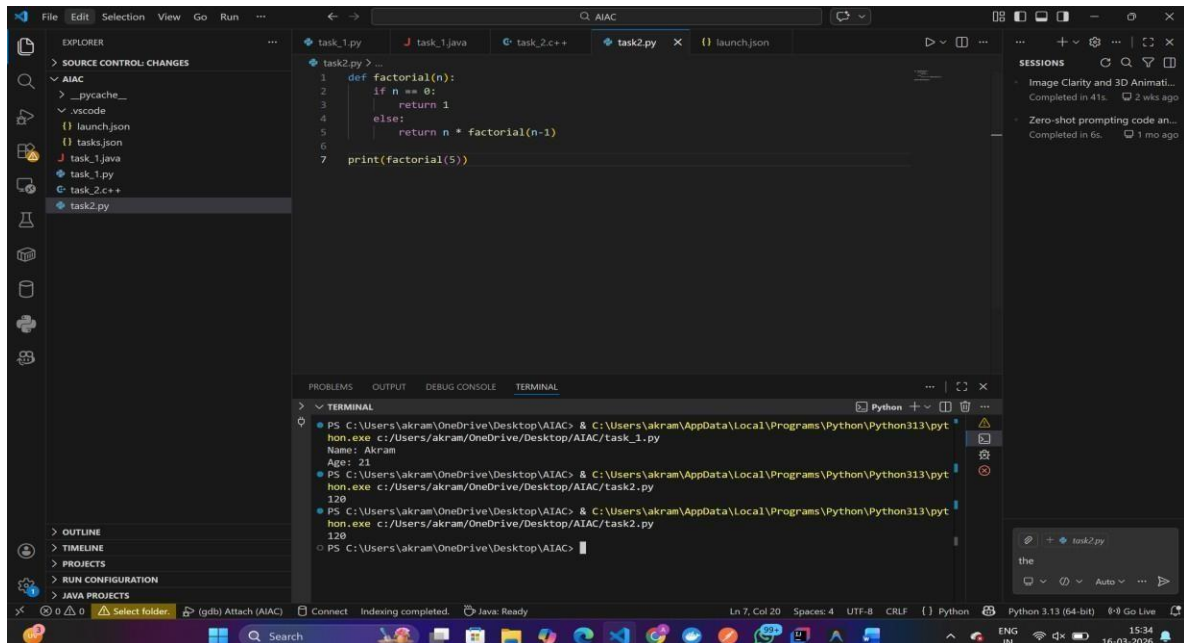
- Ask AI to translate a recursive C++ factorial function into Python.

- Validate correctness by testing with sample inputs.
- Ensure Python version follows Pythonic conventions.

Expected Output:

- A correct Python implementation equivalent to the original C++ factorial function.

## CODE:



The screenshot shows the Visual Studio Code editor with a Python file named `task2.py` open. The code implements a recursive factorial function. The Explorer sidebar on the left shows the project structure with files `task1.py`, `task1.java`, `task2.c++`, and `task2.py`. The Terminal at the bottom shows the execution of the Python script, which outputs `Age: 21` and `120`.

```

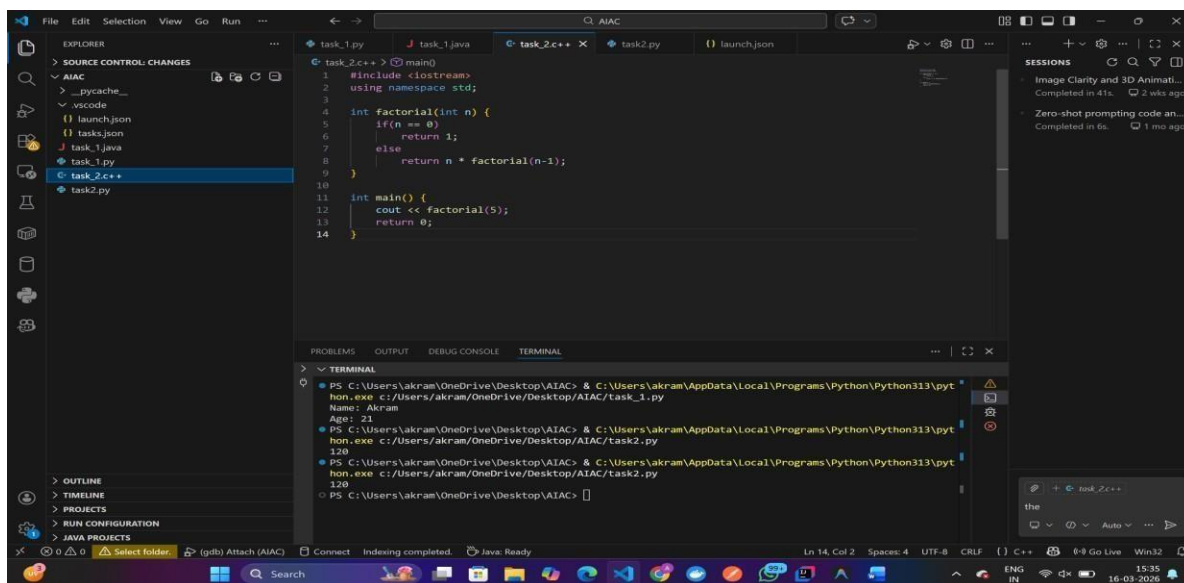
1 def factorial(n):
2     if n == 0:
3         return 1
4     else:
5         return n * factorial(n-1)
6
7 print(factorial(5))

```

```

PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:/Users/akram/OneDrive/Desktop/AIAC/task_1.py
Name: Akram
Age: 21
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:/Users/akram/OneDrive/Desktop/AIAC/task2.py
120
PS C:\Users\akram\OneDrive\Desktop\AIAC>

```



The screenshot shows the Visual Studio Code editor with a C++ file named `task2.c++` open. The code implements a recursive factorial function. The Explorer sidebar on the left shows the project structure with files `task1.py`, `task1.java`, `task2.c++`, and `task2.py`. The Terminal at the bottom shows the execution of the C++ program, which outputs `Age: 21` and `120`.

```

1 #include <iostream>
2 using namespace std;
3
4 int factorial(int n) {
5     if(n == 0)
6         return 1;
7     else
8         return n * factorial(n-1);
9 }
10
11 int main() {
12     cout << factorial(5);
13     return 0;
14 }

```

```

PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:/Users/akram/OneDrive/Desktop/AIAC/task_1.py
Name: Akram
Age: 21
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:/Users/akram/OneDrive/Desktop/AIAC/task2.py
120
PS C:\Users\akram\OneDrive\Desktop\AIAC>

```

## Task 3 – SQL to Pandas Query

Task:

Translate a SQL query into a Pandas DataFrame operation in Python.

Instructions:

- Provide AI with a SQL query (e.g., `SELECT name, salary FROM employees WHERE salary > 50000`).
- Ask AI to generate equivalent Pandas code.
- Test the generated code on a sample DataFrame.

Expected Output:

- Equivalent Pandas query that retrieves the correct subset of data.

CODE:

The screenshot shows the VS Code editor with a Python file named `task_3.py` open. The code defines a dictionary `data` with employee names and salaries, creates a `pandas` `DataFrame`, and filters for employees with a salary greater than 50,000. The terminal shows the command `python task_3.py` being executed, followed by the output of the script, which is a `DataFrame` containing two rows: Alice (60000) and Bob (75000).

```
task_3.py > ...
1 import pandas as pd
2
3 data = {
4     'name': ['John', 'Alice', 'Bob', 'Emma'],
5     'salary': [45000, 60000, 75000, 40000]
6 }
7
8 df = pd.DataFrame(data)
9
10 result = df[df['salary'] > 50000][['name', 'salary']]
11
12 print(result)
```

```
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:\Users\akram\OneDrive\Desktop\AIAC\task_3.py
Age: 21
120
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:\Users\akram\OneDrive\Desktop\AIAC\task_2.py
120
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:\Users\akram\OneDrive\Desktop\AIAC\task_3.py
   name  salary
2  Alice  60000
3    Bob  75000
PS C:\Users\akram\OneDrive\Desktop\AIAC>
```

The screenshot shows the VS Code editor with a SQL file named `task_3.sql` open. The code contains a `SELECT` query that filters for employees with a salary greater than 50,000. The terminal shows the command `python task_3.py` being executed, followed by the output of the script, which is a `DataFrame` containing two rows: Alice (60000) and Bob (75000).

```
task_3.sql
1 SELECT name, salary
2 FROM employees
3 WHERE salary > 50000;
```

```
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:\Users\akram\OneDrive\Desktop\AIAC\task_1.py
Age: 21
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:\Users\akram\OneDrive\Desktop\AIAC\task_2.py
120
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:\Users\akram\OneDrive\Desktop\AIAC\task_3.py
   name  salary
2  Alice  60000
3    Bob  75000
PS C:\Users\akram\OneDrive\Desktop\AIAC>
```

## Task 4 – Real-Time Application: Multi-Language Snippet

Converter Scenario:

Students will choose a real-time use case where they need code in multiple languages. Example:

- Converting a Python web scraping snippet into JavaScript for browser automation.

- Translating a Java method into C# for enterprise applications.

Instructions:

- Prompt AI to translate code between two chosen languages.
- Test both implementations to ensure they produce the same results.
- Document challenges faced during translation (e.g., syntax, libraries).

Expected Output:

- Working equivalent code snippets in at least two different programming languages

CODE:

The screenshot shows a Visual Studio Code editor window with a project named 'AIAC'. The Explorer sidebar on the left shows a file tree with files like 'task\_1.py', 'task\_2.c++', 'task\_3.py', 'task\_3.sql', 'task\_4.py', and 'task2.py'. The main editor area displays the content of 'task\_4.py', which is a Python script using the 'requests' and 'BeautifulSoup' libraries to fetch data from a URL and print the href of the first link. The script is as follows:

```
1 import requests
2 from bs4 import BeautifulSoup
3
4 url = "https://www.onlinegdb.com/"
5
6 response = requests.get(url)
7 soup = BeautifulSoup(response.text, 'html.parser')
8 for link in soup.find_all('a'):
9     print(link.get('href'))
10
```

Below the editor, the TERMINAL panel shows the command prompt output. The command executed is 'python task\_4.py' from the directory 'C:\Users\akram\OneDrive\Desktop\AIAC'. The output shows the script running successfully and printing the href of the first link found on the page, which is 'https://www.onlinegdb.com/blog/brief-guide-on-how-to-use-onlinegdb-debugger/'.

```
PS C:\Users\akram\OneDrive\Desktop\AIAC> python task_4.py
https://www.onlinegdb.com/blog/brief-guide-on-how-to-use-onlinegdb-debugger/
```

